LGCANet: Local–Global and Change-Aware Network via Segment Anything Model for Remote Sensing Images Change Detection

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***Abstract*— Change detection (CD) is a very fundamental and challenging task in remote sensing. Many deep learning-based CD methods generally utilize Siamese networks to extract image features. However, the semantic features extracted by these methods are still not fine-grained. In addition, these CD methods ignore the object scale diverse in remote sensing images and the interaction information between bitemporal images, which leads to the problem that the network is unable to capture more efficient feature embeddings, with ambiguous or erroneous detection results. To alleviate the above issues, we propose local–global and change-aware network (LGCANet) via fast segment anything model (SAM). The SAM can accurately segment objects in various scene images. In this work, we intend to utilize the powerful recognition capabilities of SAM to refine the CD task. Therefore, LGCANet employs more efficient FastSAM and ResNet as encoders to extract potential feature representations in remote sensing images. FastSAM can effectively extract global contextual information, combined with ResNet's powerful deep feature extraction capability, which enables the network to comprehensively model features. LGCANet contains three modules: content-aware attention module (CAAM), fore–background-aware module (FAM), and edge-reinforce hybrid-selection module (EHM). CAAM delivers feature extraction from local to global perception, realizing dynamic attention to various scales of objects. FAM can effectively learn foreground and background representations through feature interaction, which significantly enhances the model's capability of recognizing changed regions. EHM can utilize direction-awareness to extract edge information and generate fine-grained detection maps by adaptively selecting discriminative features through designed attention mechanisms. Experiments on publicly available CD datasets show that LGCANet achieves superior detection performance compared to other state-of-the-art methods. The code is available at https://github.com/Jscript10/LGCANet**

***Index Terms*— Change detection (CD), content-aware attention module (CAAM), edge-reinforce hybrid-selection module (EHM), fore–background-aware module (FAM), remote sensing, segment anything model (SAM).**

I. Introduction

CHANGE detection (CD) is an important research topic in the field of remote sensing (RS) and computer vision, which aims to identify and analyze changes that occur in same geographical area between different time points. Due to the rapid development of human society, CD techniques have been widely applied in the fields of urban expansion [1], land cover [2] and agricultural monitoring [3].

Early CD methods usually can be categorized into algebraic operations [4], image transformations [5], and imagery classification [6], which provide diversified development paths for CD task. Nevertheless, these approaches are mainly intended for low- or medium-resolution remote sensing image, which cannot satisfy the task requirements of current large-scale very high-resolution (VHR) CD datasets. In addition, traditional CD methods are highly dependent on manual-designed features, which may failed to capture the complex change patterns and are sensitive to noise, thus failing to generate fine-grained detection results.

Recently, the emergence of large-scale data and increasing computational power have provided favorable conditions for deep learning techniques. Consequently, numerous neural network models have been implemented for diverse domains including CD [7]. Most popular deep learning-based CD methods utilize the Siamese network structure, which obtains bitemporal image characteristics via a weight-sharing ResNet [8], followed by a decoder that generates changed regions. For instance, Daudt et al. [9] introduced three CD networks (FC-EF, FC-Diff, and FC-Conc). Although above networks show better detection performance, they still fail to extract discriminative features. Fang et al. [10] proposed Siamese-based network and UNet++ (SNUNet), which extracts multiscale variations by integrating channel attention. To increase the network's attention to the spatial relations of bitemporal images, Chen and Shi [11] introduced spatial-temporal attention network (STANet), which extracts rich feature representations through the spatial self-attention mechanism. Peng et al. [12] presented differential enhancement densely attention convolutional neural net (DDCNN), which takes differential images as weights to guide DDCNN extract the change regions. Moreover, transformer-based CD models (BIT [13], ChangeFormer [14]) use semantic tokens to encode bitemporal image features, then fused with depth features to produce a change map.

Although the above methods have achieved excellent performance, they still suffer from several problems. First, as shown in Fig. 1, ResNet-based backbone network has an inherent problem that network is incapable of extracting discriminative features during the encoding phase, which leads to the network

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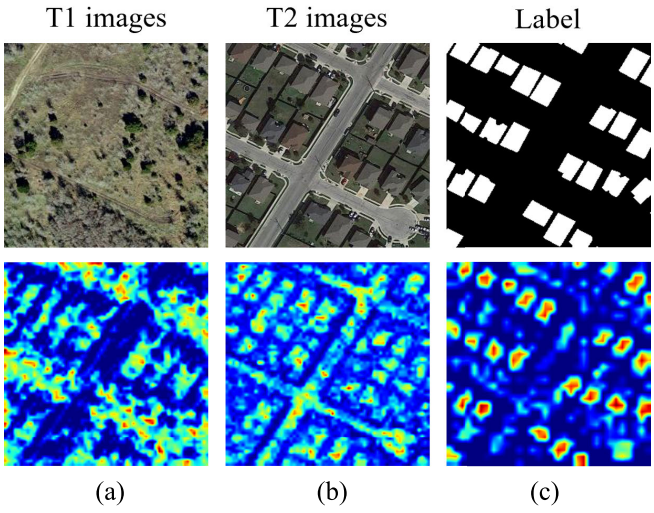


Fig. 1. Illustration of network visualization. (a) ResNet extracts features with more noise. (b) FastSAM extracts more fine-grained results, but with redundant features, e.g., road features. (c) Our approach can extract more fine-grained and accurate change features.

only extracting changed regions in the decoding phase, thus easily leading the network focusing on background noise. Second, they have not deeply investigated the object scale diversity of remote sensing images, e.g., SNUNet focuses on extracting local information and BIT is mainly utilized to capture global semantic information; this design restricts the network's ability to discriminate complex targets. In addition, the current methods, e.g., DDCNN, simply utilize difference images or attention mechanisms to extract foreground features, with no consideration of network learning both foreground and background features during the encoding phase. The background feature learning helps the network to better understand the overall structure of the image, which can further improve the model's ability to recognize changed regions. Moreover, the above methods suffer from different degrees of incomplete and blurred detection edges, which is mainly caused by the network's failure to sufficiently refine high-quality semantic information in decoding stage, resulting in insufficient expressive ability of edge details. It remains a challenge to generate complete and detailed edge information at the decoding stage. Therefore, it is essential to design a novel CD network to address these issues.

Recently, segment anything model (SAM) [15] and its variant FastSAM [16] have been extensively utilized in computer vision field, and it has demonstrated powerful segmentation capabilities in image segmentation, detection, and scene understanding. For instance, TPP-SAM [17] achieves zero-shot road extraction from high-resolution remote sensing imagery through trajectory point prompting. PointSAM [18] employs prototype-based regularization and negative prompt calibration strategies to enhance segmentation performance under point supervision. In this article, we want to improve the CD task by using FastSAM to improve the encoder's feature extraction capability. However, it might have limited generalization ability for direct application in remote sensing because its training dataset contains large number of natural images. For instance, SAM-CD [19], which uses FastSAM as an encoder to extract image feature representations. As shown in Fig. 1, FastSAM focuses on the image with regular objects and will extract redundant features such as roads and other features.

To address these issues, we intend to utilize the segmentation capability of FastSAM and the feature extraction capability of ResNet [8] to improve CD task, thus increasing the network training efficiency and detection accuracy. In this article, we propose local–global and change-aware network (LGCANet) via fast SAM. LGCANet consists with three modules: content-aware attention module (CAAM), fore–background-aware module (FAM), and edge-reinforce hybrid-selection module (EHM). Overall, this article's contributions are summarized below.

1) We design a novel CD network LGCANet, which combines FastSAM and ResNet as encoders. We enhance the network's feature representation capability by fusing the features of two branches at the encoding stage, thus capturing the semantic information of objects more efficiently.
2) We present CAAM that can capture channel and spatial semantic features from local to global perspectives, enabling feature extraction for objects at different scales. Moreover, CAAM employs learnable offsets and weights to adaptively fuse different types of features, providing richer contextual information and detail capturing capabilities.
3) To better allow the network to learn foreground and background feature representations, we propose FAM. FAM promotes the network's learning of changed and unchanged regions by generating different types of mask mappings, which allows the network to enhance its attention to foreground features and suppress the interference of contextual noise.
4) We present EHM to enhance the edge information of the extracted feature embedding. It enhances the details of changed regions through direction awareness and enhances the contextual interaction of features through channel shuffling and hybrid selection mechanisms to generate more robust detection results.

The following section of this article is arranged as follows. Section II overviews the relevant work on CD task and SAM. Section III demonstrates our proposed LGCANet. Detailed experimental outcomes and analysis are presented in Section IV. Last, Section V summarizes our work.

II. Related Work

A. Visual Foundation Models (VFMs)

VFMs enable models to extract feature knowledge from large-scale image data and apply it to various downstream tasks, such as image segmentation and instance segmentation. In computer vision, convolutional neural networks (CNNs) are generally utilized to extract image features and has become the mainstream framework. ResNet [8] was proposed to solve the neural network gradient vanishing problem and implement scalable deep learning model. Furthermore, many effective network architectures have been proposed, such as ResNeXt [20] and EfficientNet [21]. Recently, ViT [22] has

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achieved great success in visual tasks due to its global perceptual field. Transformer-based vision models have pushed the field forward, and Zhai et al. [23] extended ViT to 2 billion parameters and it significantly improved the performance of VFMs. Contrastive language–image pretraining (CLIP) [24] uses large-scale data (400 million text–image pairs) for training, which uses contrast-based learning to empower the model learning text–image matching relations, enhancing model generalization to multimodal domains.

Currently, the computer vision field has sparked a trend of exploring the foundation models due to the presentation of SAM [15]. SAM generates high-quality image segmentation results from input prompts, which are trained on datasets containing 1.1 billion masks and more than 11 million images, with strong zero-sample learning performance. Later, Li et al. [25] proposed OMG-Seg, which employs a shared decoder to handle all types of queries and features, realizing cross-task parameter sharing. OMG-Seg can process up to ten different segmentation missions, covering image segmentation, semantic segmentation, and video object segmentation. Zhao et al. [16] proposed an improved version of SAM named FastSAM, which accelerates the processing speed of image segmentation while maintaining high accuracy. FastSAM achieves more efficient performance by optimizing model architecture and reducing computational complexity.

Although SAM and its variants have shown strong performance in downstream tasks, they do not show superior results in specific domains, e.g., medical images and remote sensing images. It is because their training data are built on large-scale natural images, they cannot capture objects with very small and irregular scales in remote sensing images properly. In this work, we consider combining FastSAM with CNN to learn potential semantic mappings in remote sensing maps.

B. Deep Learning-Based CD Approaches

With the excellent performance of deep learning in vision missions, deep learning-based CD approaches have been rapidly evolved in recent years, and a variety of effective network architectures have emerged. Fully convolutional networks (FCNs) [26] is a type of end-to-end learning approach, which can directly predict pixel-level change information from input image. However, FCN may be affected by data noise and complex environments when processing remote sensing images, resulting in insufficient accuracy. It is due to the fact that the network input is stitched from bitemporal images, which causes the network fail to capture small target variations in remote sensing images [27].

To adapt the input for CD task, Daudt et al. [9] proposed Siamese-based CD networks, named FC-Diff and FC-Conc, respectively. It enhances the network's ability to detect changed regions by introducing weight-shared encoders to process the bitemporal image. Later, SRCDNet [28] introduces hybrid attention mechanisms to enhance the details of changed regions for reducing contextual noise. Li et al. [29] proposed densely attentive refinement network (DARNet), which combines dense skip-connection and hybrid attention modules to progressively refine the predicted change results. Wang et al. [30] proposed attention-aware sobel GCN (ASGCN), which introduces graph convolutional networks to learn the connection and interaction information between feature nodes, thus capturing the changed region between bitemporal images. Li et al. [31] proposed interactive and dynamic-aware Siamese network (IDA-SiamNet), which enhances the extraction of changed regions by feature interaction in spatial and channel perspectives. Chen et al. [32] proposed ChangeMamba, which introduces the state space-based Mamba architecture into remote sensing CD task to extract accurate change information. In addition, BiT [33] introduced transformer architecture to tackle CD task, it allows capturing long-range dependency and global contextual information in remotely sensed images. SARASNet [34] strengthens the character extraction ability of network by combining CNN with Transformer and uses spatial and scale-aware blocks to address the boundaries noise produced by different scale objects, and to augment the feature representation of interaction details.

Although the above methods achieve better detection performance, they still have weaknesses for the extraction of scale and object shape diverse features in remote sensing images. In addition, it remains a challenge to make the network effectively learn variation features and generate fine-grained detection edges.

III. Methodology

A. Overview

The proposed LGCANet architecture is shown in Fig. 2. In the encoding phase, we use FastSAM's encoder and ResNet-18 to extract discriminative semantic features. Assume x_1, x_2 are input bitemporal images, $F(\cdot)$ and $R(\cdot)$ denote FastSAM and ResNet feature extractors, respectively. The encoding procedure can be denoted below

$$\begin{aligned} F &= \alpha \cdot F(x_1, x_2) + \beta \cdot R(x_1, x_2) \\ F &= \left\{ f_t^1, f_t^2, f_t^3, f_t^4 \right\}, \quad t = 1, 2 \end{aligned} \tag{1}$$

where α and β are learnable weight parameters which can automatically align the weights during the training process to optimize the features of different extractors. f indicates the extracted feature map and t indicates moment. We perform feature aggregation from four different scales, denoted as $f_t^1, f_t^2, f_t^3, f_t^4$. After that, we perform channel reduction by convolution, which can be represented as below

$$F' = \text{relu}(\text{bn}(\text{Conv}(F))) \tag{2}$$

where Conv is a convolution with core size 1×1, bn indicates batch normalization function, and relu is the rectified linear unit (ReLU) operation.

To facilitate the network's learning of changed regions, we extracted bitemporal image semantic characteristics to assist the generation of changed regions. The equation is indicated as below

$$\begin{aligned} F'' &= \text{relu}\left(\text{bn}\left(\text{Conv}\left(F'\right)\right)\right) \\ s_1, s_2 &= \text{Linear}\left(F''\right) \end{aligned} \tag{3}$$

where Linear denotes linear embedding operation, and s_1 and s_2 denote the generated semantic features from multitemporal

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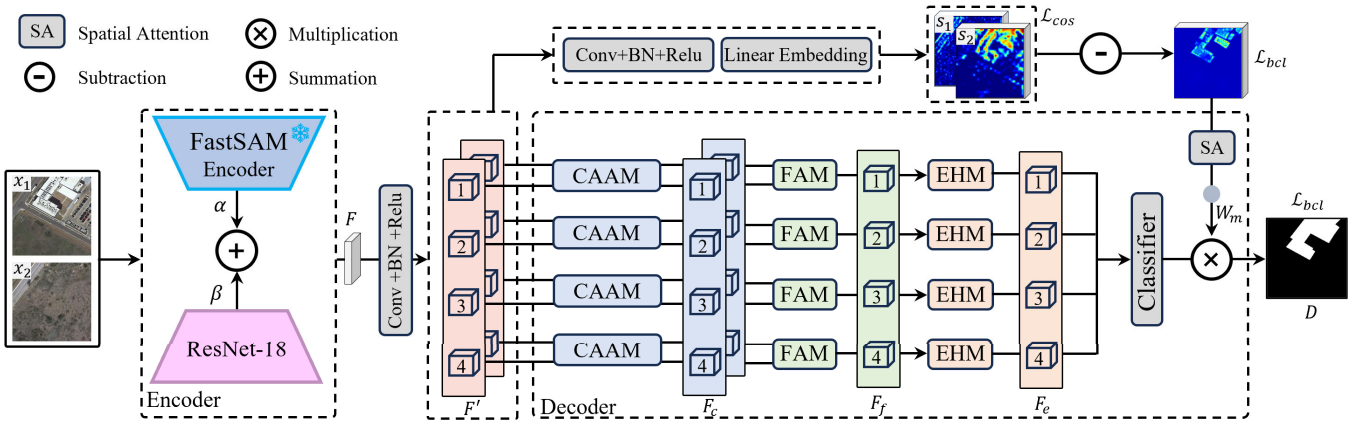


Fig. 2. Overall structure of LGCANet.

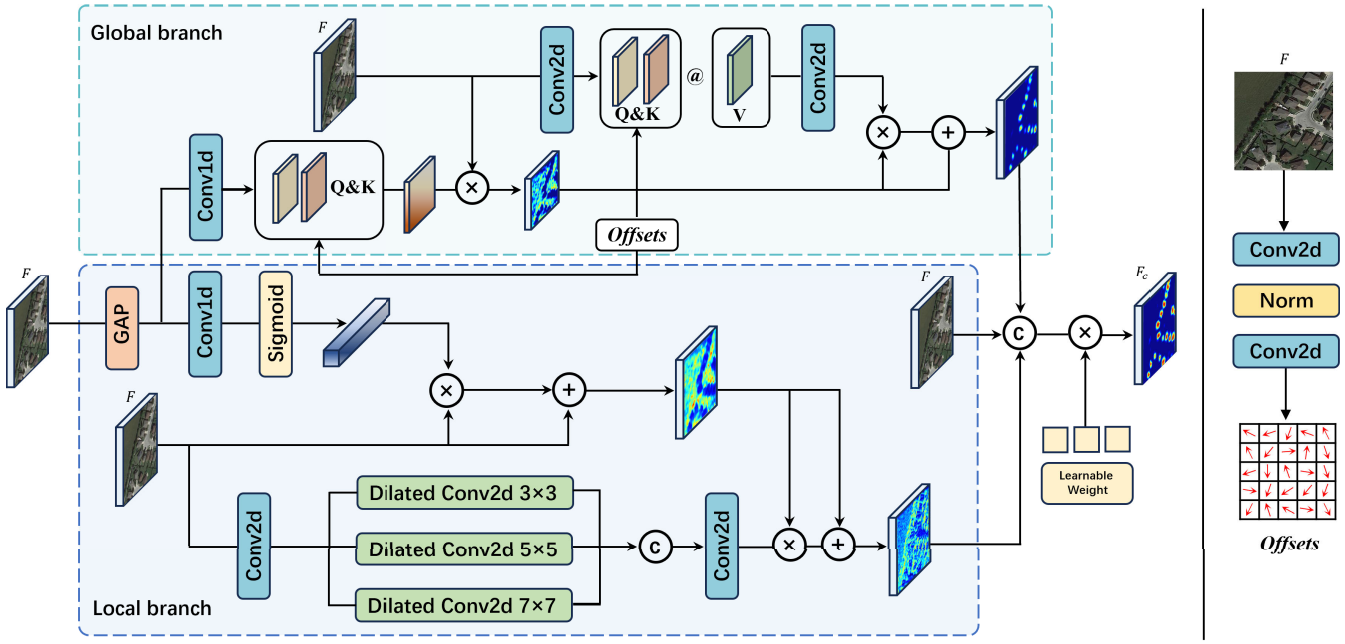


Fig. 3. Illustration of CAAM.

images. We make difference of the semantic features to get the feature m, then m passes through the spatial attention to get the attention weight W_m, and the process is denoted as follows:

$$m = |s_1 - s_2|$$
$$W_m = \sigma(\text{MLP}(m)) \tag{4}$$

where $-$ indicates the differential processing, MLP means the multilayer perception, and σ stands for sigmoid activation function. After that, W_m is applied as weights to drive the networks to attention the changing semantic features.

Meanwhile, F' enters the CAAM module to get the feature F_c. CAAM is used to capture multiscale, various shapes of object feature in remote sensing images. After that, f_c goes to FAM to get F_f, and FAM interacts foreground features with background features. This enables networks to be attentive on changes while mitigating the effects of contextual noise. To extract the contour information of the features, F_f enters the EHM module for edge extraction and feature selection to generate smooth feature F_e. After that, F_e undergoes feature fusion and channel dimensionality reduction by the Classifier to generate the binary graph D'. Finally, the weights W_m generated by semantic mapping are utilized to supervise the production of change map D, with the below equation

$$D' = \text{Classifier}(F_e)$$
$$D = W_m \times D' \tag{5}$$

and finally, we optimize the loss by s_1, s_2, m, and D for both changed and unchanged regions. The detailed formulations are described in Section III-E.

B. Content-Aware Attention Module

We designed CAAM to capture multiscale irregular objects, as illustrated in Fig. 3. It includes two branch, the local and global branches. Local branch captures contextual information from the image to obtain local channel attention and local spatial attention. Global branch allows channel and spatial interactions and further enhances the modeling capability, via introduces learnable offsets for additional dynamic positional biases. The feature mappings generated by two branches are

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merged to generate local–global features, later aggregated into global feature representations by adaptive weighting.

1) Local Branch: Using the feature F of $c \times h \times w$ as example, the feature compresses the spatial information by global average pooling (GAP) to obtain a tensor of size $c \times 1 \times 1$. Later, the interchannel interactions are realized by a 1-D convolution with core size of 3×3. Sigmoid activation function is utilized to capture the local channel attentional feature W_l^c. Calculated as below

$$W_l^c = \sigma(\text{Conv1}d(\text{GAP}(F))) \tag{6}$$

to capture the multiscale objects, we perform feature dimension reduction on input maps F by convolution of 1×1, which serves for removing redundant information. Later, local spatial context features are extracted by dilated convolution with receptive field sizes of 3×3, 5×5, and 7×7. Dilated convolution increases the receptive fields with no computational increase, allowing the model to extract complex and multiscale feature representations more efficiently. Subsequently, features are channel concatenated and dimensionality recovery is realized by 1×1 convolution. The formula is given below

$$\begin{aligned} F_r &= \text{Conv2}d(F) \\ F_{\text{dc}}^3 &= D\text{Conv2}d_{3\times3}(F_r) \\ F_{\text{dc}}^5 &= D\text{Conv2}d_{5\times5}(F_r) \\ F_{\text{dc}}^7 &= D\text{Conv2}d_{7\times7}(F_r) \\ W_l^s &= \text{Conv2}d\left(\text{Cat}\left(F_{\text{dc}}^3, F_{\text{dc}}^5, F_{\text{dc}}^7\right)\right). \end{aligned} \tag{7}$$

2) Global Branch: Global branch is utilized to capture the global interchannel and interpixel similarity matrices to balance feature embedding, as presented in Fig. 3. Initially, feature map F is compressed in spatial dimensions by GAP operation, followed by 1-D convolution and sigmoid activation function to obtain query tensor Q_g^c and the key tensor K_g^c. Unlike the ordinary self-attention mechanism, we introduce an offset matrix in the global branch, which is used to achieve more efficient information aggregation. Embedding the offset matrix into query (Q) and key (K) tensors, thus allowing the offset matrix to interact with Q and K tensor for additional dynamic changes. Calculations are as follows:

$$\begin{aligned} Q_g^c, K_g^c &= \text{Conv1}d(\text{GAP}(F)) \\ \text{offsets}'_{(i,j)} &= \text{relu}(\text{Conv2}d(F)) \\ \text{offsets}_{(i,j)} &= k \cdot \tanh\left(\text{Conv2}d\left(\text{offsets}'_{(i,j)}\right)\right) \end{aligned} \tag{8}$$

where the first convolution is utilized to reduce the channel count to $(c/8)$, while relu activation functions are implemented. Nonlinear transformations are applied to enable the model to capture complex features and relationships. The second convolution further reduces the channel to 2, corresponding to the offsets in the x- and y-directions. Later, tanh function is used to constrain the offset matrix to be between -1 and 1, and then multiplied by k to expand the offset to vary within a reasonable range to prevent invalid sampling, which is set to 6 in this experiments. Finally, query and key tensor performs outer product operation, and Softmax function is operated on the channels to obtain the global channel similarity matrix W_g^c with size $c \times c$. The global channel similarity matrix is multiplied by input feature F to acquire the global channel attention map F_g^c. The formula is given below

$$\begin{aligned} \text{new}Q_g^c(i,j), \text{new}K_g^c(i,j) &= \left(Q_g^c(i,j) + \text{offset}(i,j)\right) \\ &\quad \left(K_g^c(i,j) + \text{offset}(i,j)\right) \end{aligned} \tag{9}$$

$$\begin{aligned} Q_g^{co}(i,j) &= \left(2 \cdot \frac{\text{new}Q_g^c(i,j)^x}{W-1} - 1, 2 \cdot \frac{\text{new}Q_g^c(i,j)^y}{H-1} - 1\right) \\ K_g^{co}(i,j) &= \left(2 \cdot \frac{\text{new}K_g^c(i,j)^x}{W-1} - 1, 2 \cdot \frac{\text{new}K_g^c(i,j)^y}{H-1} - 1\right) \\ W_g^c &= \text{Softmax}\left(Q_g^{co}(i,j), K_g^{co}(i,j)\right) \\ F_g^c &= W_g^c \cdot F \end{aligned} \tag{10}$$

where $\text{new}Q_g^c(i,j)$, $\text{new}K_g^c(i,j)$ denotes the tensor after adding the offset, and i,j denote the feature coordinates in x- and y-directions. $Q_g^{co}(i,j)$, $K_g^{co}(i,j)$ is tensor after normalization, and the sampling positions are normalized to a range from -1 to 1 for subsequent learning. Then, to capture the global spatial relations, we process the feature map F with convolution of kernel size 1×1 to compress the spatial dimensions to hw, obtaining query (Q_g^s), key (K_g^s), and value (V_g^s). Similar to the global channel attention, we introduce offset matrices for Q_g^s and K_g^s, respectively, to obtain additional spatial offsets $Q_g^{so}(i,j)$, $K_g^{so}(i,j)$. Later, the matrix outer product is applied to obtain the similarity matrix, and then Softmax activation function is performed to obtain the global spatial attention map W_g^s. Then, W_g^s is multiplied by V_g^s to obtain the postinteraction features, and then channel recovery is performed with convolution. Finally, global spatial attention map M_g^s are utilized to weight F_g^c to get global attention feature map F_g. The calculations are as follows:

$$\begin{aligned} W_g^s &= \text{Softmax}\left(Q_g^{so} K_g^{so}\right) \\ M_g^s &= \text{Conv2}d\left(W_g^s \odot V_g^s\right) \\ F_g &= M_g^s \cdot F_g^c + F_g^c \end{aligned} \tag{11}$$

as depicted in Fig. 3, we concatenate features F_l and F_g obtained from the two branches with input feature F. Then, we introduce learnable feature embeddings to weight the different features and achieve the optimal fusion result. The formula is as follows:

$$F_c = \text{Att}_1 \cdot F_l + \text{Att}_2 \cdot F_g + \text{Att}_3 \cdot F. \tag{12}$$

C. Fore–Background-Aware Module

To improve the model's learning capability for changed and unchanged features, we propose FAM. FAM achieves multilevel aggregation and perception of feature maps and empowers the network's capability to extract changed regions. As shown in Fig. 4, take input maps F_1 and F_2 as example, which represent the feature of moment 1 and moment 2, respectively. Initially, the input features are differenced to produce the differential feature D. Input features are concatenated with feature D to acquire F_c. F_c is pass through convolution with kernel size of 1 for channel compression, and two tensors of $c \times 1 \times h \times w$ are obtained by applying average pooling and maximum pooling operation. After that, tensors are merged into convolution with core of 3×3 and sigmoid activation

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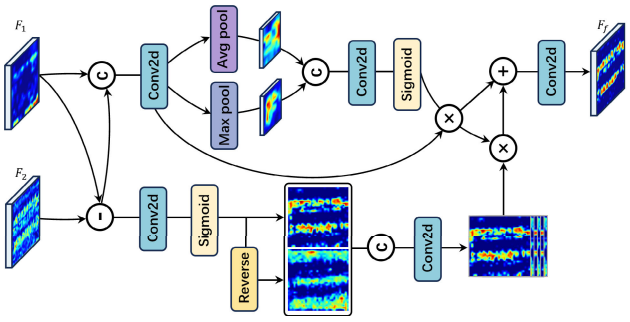


Fig. 4. Architecture of FAM.

operation to generate spatial relational attention weights A_s. It is computed as below

$$\begin{aligned} D &= |F_1 - F_2| \\ F_c &= \mathrm{Cat}(F_1, F_2, D) \\ F_c' &= \mathrm{Conv2}d_{1\times1}(F_c) \\ A_s &= \sigma\left(\mathrm{Conv2}d_{3\times3}\left(\mathrm{Cat}\left(\mathrm{Avg}\left(F_c'\right), \mathrm{Max}\left(F_c'\right)\right)\right)\right) \end{aligned} \tag{13}$$

where F_c' denotes the compressed tensor, and Avg and Max indicate the average and maximum pooling operation. Meanwhile, for difference feature D, we adopt convolution operation and sigmoid activation function to produce the variation map D_f. Then, we obtain the reverse mapping feature D_b by inversion operation. D_f and D_b describe the foreground and background information of changed features, respectively, then they are channel concatenated and processed through convolution for channel recovery to obtain the feature D_c, which contains the change mask and the unchanged mask. The calculation process is as follows:

$$\begin{aligned} D_f &= \sigma\left(\mathrm{Conv2}d_{1\times1}(D)\right) \\ D_b &= E - D_f \\ D_c &= \mathrm{Conv2}d_{1\times1}\left(\mathrm{Cat}\left(D_f, D_b\right)\right) \end{aligned} \tag{14}$$

where the value of D_f has been limited by the activation function between 0 and 1, and E denotes the unitary matrix with value 1 for realizing the eigenvalue inversion. Later, F_c' are sequentially weighted for augmented perception of spatial and difference information by using A_s and D_c. Finally, multiplayer convolution is applied to nonlinearly transform to get output feature F_f. The formula is given below

$$\begin{aligned} D_{\mathrm{out}}' &= \left(A_s \cdot F_c'\right) \cdot D_c + \left(A_s \cdot F_c'\right) \\ F_f &= \mathrm{relu}\left(\mathrm{bn}\left(\mathrm{Conv2}d_{1\times1}\left(\mathrm{Conv2}d_{3\times3}\left(D_{\mathrm{out}}'\right)\right)\right)\right). \end{aligned} \tag{15}$$

Changed masks only provide information about foreground context, and unchanged masks only provide background context feature. Neglecting any of these masks will lose the full understanding of overall image and lead to incomplete feature extraction. Through FAM, the network can generate comprehensive and detailed feature representations. Network can better comprehend where exactly change occurs and the surrounding contextual information, thus improving the discriminative power of models.

D. Edge-Reinforce Hybrid-Selection Module

To achieve edge refinement and accurate discrimination of changed regions, we propose EHM. As illustrated in Fig. 5, taking the input F_f for instance, features enter the directional convolution. Directional convolution is categorized into four directions, positive diagonal (p), vertical (v), horizontal (h), and anti-diagonal (a), which are assigned to exploit the edge information of features. Subsequently, the output is concatenated and summed with input feature to get feature F_{dir}'. The calculation formula is given below

$$\begin{aligned} F_{\mathrm{dir}} &= \mathrm{Cat}\left(\mathrm{Con}_{\mathrm{p}}\left(F_f\right), \mathrm{Con}_{\mathrm{v}}\left(F_f\right), \mathrm{Con}_{\mathrm{h}}\left(F_f\right), \mathrm{Con}_{\mathrm{a}}\left(F_f\right)\right) \\ F_{\mathrm{dir}}' &= F_{\mathrm{dir}} + F_f \end{aligned} \tag{16}$$

where F_{dir} denotes the feature after directional convolution, Con_i denotes directional convolution, and we set the perception field to 5 for effective extracting feature contours. To enhance the representation of features, we use channel shuffle operations to learn the interactions between features of different orientations. We perform channel splitting and restructuring of features to obtain the shuffled feature F_{sh}. Then, we compress the information by unfold and average pooling operations to generate consecutive patches. We use multiple linear layers to extract spatial features to obtain F_{a}. Later, Softmax function is utilized to generate probabilistic weights A_{p} for each patch, which is weighted to F_{a}. The formula is as follows:

$$\begin{aligned} F_{\mathrm{sh}} &= \mathrm{Channel}_{\mathrm{shuffle}}\left(F_{\mathrm{dir}}'\right) \\ F_{\mathrm{a}} &= \mathrm{Linear}(\mathrm{Norm}(\mathrm{Linear}(\mathrm{Avg}(\mathrm{Unflod}(F_{\mathrm{sh}}))))) \\ A_{\mathrm{p}} &= \mathrm{Softmax}(F_{\mathrm{a}}) \\ F_w &= F_{\mathrm{a}} \cdot A_{\mathrm{p}} \end{aligned} \tag{17}$$

where F_w denotes the weighted features and Norm denotes the layer normalization operation. After that, we use feature selection to extract relevant information related to the changed region. The equation is as below

$$F_e = \lambda\left(\mathrm{Clamp}_{(0,1)}(\mathrm{Cos}(F_w, \theta))\right) \tag{18}$$

where θ is a learnable feature parameter embedding that identifies the selection of those tokens that are associated with changes. Cos denotes the computation of the cosine similarity between F_e and θ. $\mathrm{Clamp}_{(0,1)}$ denotes truncation operation on the computed cosine similarity, limiting the value to between 0 and 1. λ denotes the specific channel parameter embedding, which is used to perform channel selection for each token, and finally obtain output features.

E. Optimization

To better utilize the semantic representations generated by encoders, we utilize semantic features to complement the learning of CD task. s_1 and s_2 denote the semantic features, which are first normalized to obtain a probability distribution

$$S_i = \frac{\exp(s_i)}{\sum_{i=1}^{n} \exp(s_i)} \tag{19}$$

where S_i is the probability distribution after normalization, to make the semantic features of s_1 and s_2 as similar as possible in the unchanged region, we use Boolean and logical inverse operations to get the labels of the unchanged region. The formula is as follows:

$$M_{\mathrm{unc}} = \mathrm{float}(\sim (\mathrm{bool}(M))) \tag{20}$$

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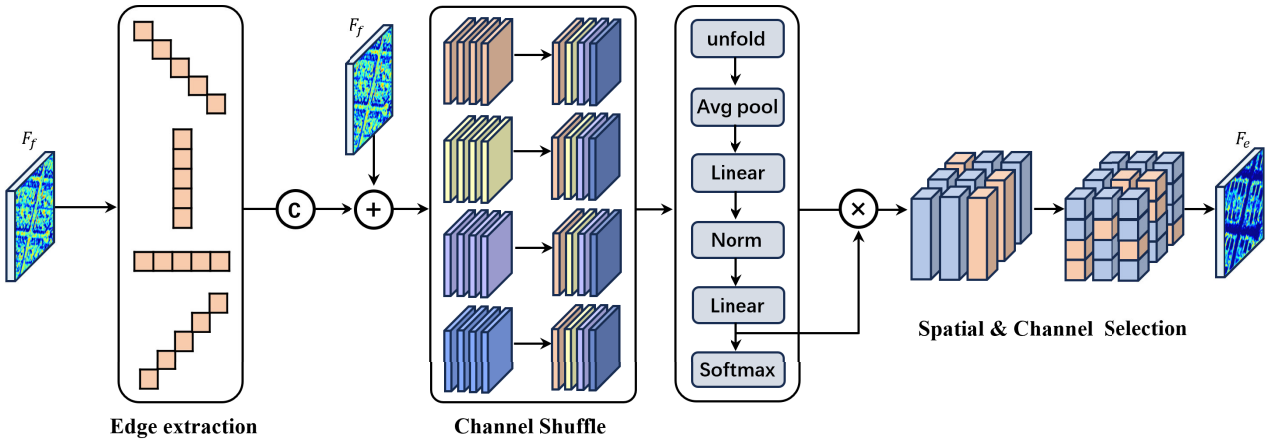


Fig. 5. Illustration of EHM.

where $\sim$ denotes logical inverse operation. Unchanged pixel value in M_{unc} is 1 and changed region into 0. Later on, we apply cosine embedding loss to calculate the similarity. The equation is given below

$$\mathcal{L}_{\cos}(S_1, S_2, M_{\text{unc}}) = \frac{1}{N}\sum_{i=1}^{N}\left(1 - \frac{S_1 \cdot S_2}{\|S_1\|\|S_2\|}\right) \cdot M_{\text{unc}} \quad (21)$$

where $\|\cdot\|$ denotes the ℓ_1 norm. For changed regions, we use batch balanced loss function (BCL) to optimize the network outputs D and the difference features m (generated via semantic features). The calculations are shown below

$$\mathcal{L}_{\text{bcl}}(A, M) = \frac{1}{2}\frac{1}{p_u}\sum_{b,i,j}\left(1 - M_{b,i,j}\right)A_{b,i,j} + \frac{1}{2}\frac{1}{p_c}\sum_{b,i,j} M_{b,i,j}\text{Max}\left(0, e - A_{b,i,j}\right) \quad (22)$$

where A denotes prediction result and M denotes the change label. p_u and p_c denote the counts of no-change and change pixel pairs, respectively, which are utilized to balance the effects of different categories. e stands for the threshold value that constrains the change pixel. Eventually, the total optimization equation is below

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\cos}(S_1, S_2, M_{\text{unc}}) + \mathcal{L}_{\text{bcl}}(D, M) + \mathcal{L}_{\text{bcl}}(m, M). \quad (23)$$

IV. Experiment

A. Datasets

To validate the effectiveness of LGCANet, we use three extensive change CD datasets, consisting of LEVIR-CD dataset [11], BCDD dataset [28], and SYSU-CD dataset [28]. A brief description of these datasets is given below.

1) LEVIR-CD: LEVIR-CD is a large-scale building CD dataset covering 20 various cities in USA. The dataset spans from 2002 to 2018, and photographs cover a variety of locations in Texas, USA. LEVIR-CD comprises 637 pairs of bitemporal images, labeled with a total of 31 333 changed objects, with an image size of 1024×1024 and space resolution of 0.5 m. To adapt the specific experimental equipment, we split the dataset into image pairs of 256×256 size. Finally, the number of training dataset, validation dataset, and test dataset are 7120, 1024, and 2048, respectively. Partial imagery of the LEVIR-CD are illustrated in Fig. 6.

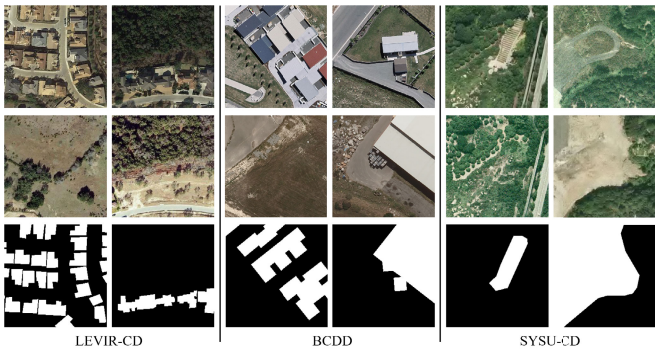


Fig. 6. Example of the dataset utilized in our article.

2) BCDD: BCDD dataset is an VHR remote sensing dataset focusing on building changes with resolution of 0.2 m and image size of 32 507 × 15 354. BCDD covers various urban environments, including building extensions, demolition, and other types. To process the data, we cropped the image to 256 × 256 size to get 7620 image pairs. Finally, the training, validation, and testing are conducted according to 8:1:1 ratio. A part of BCDD image is presented in Fig. 6.

3) SYSU-CD: SYSU-CD is a multicategory change dataset covering the time period from 2007 to 2014. The dataset is collected by Sun Yat-sen University and the locations mainly cover the Hong Kong region. The types of changes include changes in urban housing, high-rise buildings, and coastal facilities. SYSU-CD comprises 20 000 pairs of remote sensing imagery, which have size as 256×256 and resolution of 0.5 m. For deep learning applications, 12 000 pairs of images are used for training, 4000 pairs of images for validation, and 4000 pairs of images for testing. Some imagery of SYSU-CD are illustrated in Fig. 6.

B. Implementation Details

1) Evaluation Metrics: To estimate the capability of the presented approach, we utilize recall (Rec), precision (Pre),

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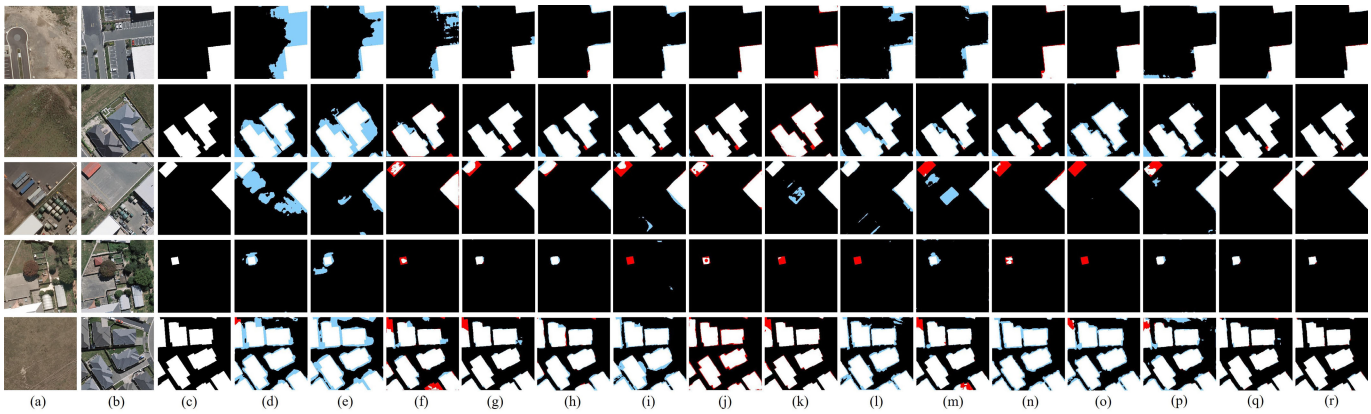


Fig. 7. Visualization on LEVIR-CD datasets. (a) T1 moment images. (b) T2 moment images. (c) Change label. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) SNUNet. (h) STANet. (i) SRCDNet. (j) BITNet. (k) DDCNN. (l) DSMANet. (m) ChangeFormer. (n) DETNet. (o) UGSNet. (p) HANet. (q) SEIFNet. (r) Our LGCANet; black represents TN, white represents TP, red represents FN, and blue represents FP.

$F1$-score, and intersection over union (IoU) for quantitative analysis. The relevant definitions are listed below

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{24}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{25}$$

$$F1 = \frac{2\text{Precision} \cdot \text{Recall}}{\text{Prcision} + \text{Recall}} \tag{26}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{27}$$

where TP and FP denotes true-positive and false-positive pixels. TN, FN denotes true-negative and false-negative pixels. For CD task, the higher $F1$-score and IoU indicate the better detection performance of method.

2) Implementation: LGCANet is implemented by Pytorch framework [35] and trained on single NVIDIA RTX 3090 GPU. For network training, we adopt the Adam optimizer [36] with a weight decay coefficient of 0.0025 and initial learning rate of $1.25e^{-4}$ with momentum of 0.9. To prevent network overfitting, we employ commonly used data augmentation strategies, such as random selection, time-swapping, and vertical flipping operations. In addition, we performed five independent random runs for both the proposed method and comparative methods, with the final accuracies calculated as the mean value across all runs to mitigate the impact of randomness.

C. Comparing Experiments

We compared with 14 state-of-the-art deep neural network-based CD approaches, including FC-EF [9], FC-Conc [9], FC-Diff [9], SNUNet [10], STANet [11], SRCDNet [27], BIT [13], DDCNN [12], DSAMNet [37], DETNet [38], UGSNet [39], ChangeFormer [14], HANet [40], and SEIFNet [41].

1) Results on LEVIR-CD Dataset: Table I presents the quantitative evaluation results of comparison approaches and LGCANet. Blue color indicates the highest score and red color indicates the second highest score. Our method achieved the top score, and then SEIFNet in second place. LGCANet achieves 89.78% and 85.32% on $F1$-score and IoU, respectively. Compared to SEIFNet, the proposed approach improves Pre and IoU on LEVIR-CD dataset by 5.14% and 3.69%, respectively.

TABLE I
COMPARATIVE EXPERIMENTS RESULTS ON LERIV-CD DATASET

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)
FC-EF	73.16	91.93	81.27	68.47
FC-Diff	74.56	93.43	82.93	70.85
FC-Conc	74.90	94.40	83.52	71.72
SNUNet	88.57	90.63	89.59	81.14
STANet	88.05	90.11	89.06	80.27
SRCDNet	83.36	90.36	86.53	76.33
BITNet	87.97	91.31	89.60	81.15
DDCNN	87.46	92.15	89.73	81.38
DSAMNet	80.44	92.22	85.90	75.29
ChangeFormer	87.86	89.95	88.89	80.02
DETNet	87.39	88.61	87.98	78.55
UGSNet	89.94	89.60	89.77	81.44
HANet	82.30	91.80	87.78	76.66
SEIFNet	88.76	91.05	89.89	81.63
Ours	93.90	87.28	89.78	85.32

Fig. 7 shows the visualization test results of various approaches on LEVIR-CD dataset. As observed from rows 1–4 in Fig. 7, LGCANet performs excellently on irregularly shaped changed regions with various scales, which successfully distinguishes the changed regions from the background and achieves the minimum omission and false detections. In addition, it can be seen from Fig. 7 row 5 that LGCANet successfully suppresses pseudochanges, while DSAMNet, BIT, HANet, SEIFNet, and other networks are unable to accurately classify pseudo-changes.

2) Results on BCDD Dataset: To validate the efficiency of LGCANet, we conducted experiments on BCDD dataset.

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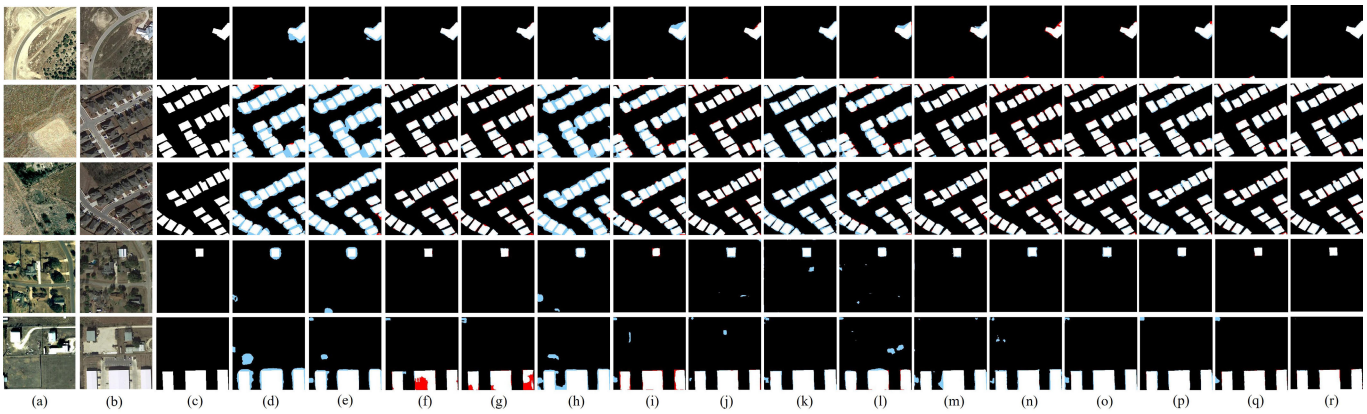


Fig. 8. Visualization on BCDD datasets. (a) T1 moment images. (b) T2 moment images. (c) Change label. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) SNUNet. (h) STANet. (i) SRCDNet. (j) BITNet. (k) DDCNN. (l) DSMANet. (m) ChangeFormer. (n) DETNet. (o) UGSNet. (p) HANet. (q) SEIFNet. (r) Our LGCANet; black represents TN, white represents TP, red represents FN, and blue represents FP.

TABLE II
COMPARATIVE EXPERIMENTS RESULTS ON BCDD DATASET

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)
FC-EF	62.04	92.40	74.23	59.03
FC-Diff	62.42	96.24	75.73	60.94
FC-Conc	78.41	86.24	82.13	69.69
SNUNet	82.85	93.50	87.85	78.34
STANet	91.99	93.26	92.62	86.26
SRCDNet	88.92	92.01	90.44	82.56
BITNet	95.20	90.64	92.86	86.69
DDCNN	96.49	87.59	91.82	84.89
DSAMNet	74.97	83.33	78.92	65.19
ChangeFormer	90.17	85.31	87.67	78.06
DETNet	90.77	90.53	90.47	82.90
UGSNet	91.59	88.01	89.76	81.42
HANet	93.94	68.34	79.12	65.46
SEIFNet	91.72	94.65	93.16	87.21
Ours	94.77	96.79	95.76	92.13

TABLE III
COMPARATIVE EXPERIMENTS RESULTS ON SYSU-CD DATASET

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)
FC-EF	73.98	76.69	75.24	60.30
FC-Diff	86.28	63.31	72.86	57.40
FC-Conc	76.14	76.81	76.43	61.92
SNUNet	75.11	80.47	77.69	63.52
STANet	61.40	88.11	72.35	56.70
SRCDNet	75.65	65.66	70.29	54.20
BITNet	77.73	78.39	77.98	63.91
DDCNN	73.19	82.09	77.06	62.74
DSAMNet	82.66	68.73	75.03	60.04
ChangeFormer	79.74	77.45	76.27	64.60
DETNet	83.06	77.37	80.12	66.83
UGSNet	81.41	81.20	81.31	68.50
HANet	76.07	72.06	73.98	58.71
SEIFNet	81.04	82.73	81.75	69.13
Ours	82.44	83.13	82.79	70.63

The quantitative test results are shown in Table II, which demonstrate the significant advantages of LGCANet in CD task. Our approach attains state-of-the-art performance of 92.13% on IoU and 95.76% on $F1$-score. Compared to current well-performed BIT and SEIFNet, the IoU is improved by 5.44% and 4.92%, and $F1$-score surpasses 2.90% and 2.60%, respectively.

Fig. 8 shows the visualization outcomes of the LGCANet and other comparison approaches on BCDD dataset. Results indicate that LGCANet successfully recognizes multiscale building changes in complex scenes. Columns 4–6 are FC-EF, FC-Diff, and FC-Conc, respectively, and they fail to accurately distinguish changed regions, leading to serious false detections and blurred edges. In addition, none of the other deep learning-based methods can detect small-scale changes, such as the variation in fourth row of Fig. 8. In contrast, LGCANet can completely detect the changed region of small targets with fewer errors and missed detections, which demonstrates the efficiency of our approach.

3) Results on SYSU-CD Dataset: Table III illustrates the quantitative experimental results of the various methodologies on SYSU-CD dataset. Our method achieves the best performance on such complicated variation dataset. LGCANet achieves Rec of 83.13%, $F1$-score of 82.79%, and IoU of 70.63%. Compared with the second best SEIFNet, our approach improves the $F1$-score and IoU by 1.04% and 1.50%.

Fig. 9 shows the visualization results on SYSU-CD. As shown from graph, LGCANet achieves the best detection results. For regular changed regions with different scales (Fig. 9, 1–3 rows), LGCANet is able to accurately capture the changed regions, suppressing pseudo-changes such

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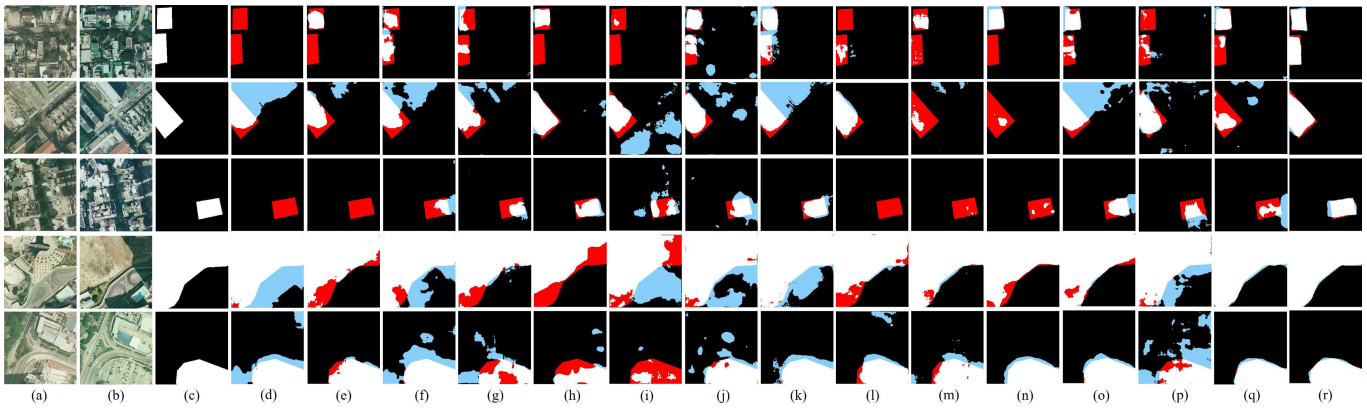


Fig. 9. Visualization on SYSU-CD datasets. (a) T1 moment images. (b) T2 moment images. (c) Change label. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) SNUNet. (h) STANet. (i) SRCDNet. (j) BITNet. (k) DDCNN. (l) DSMANet. (m) ChangeFormer. (n) DETNet. (o) UGSNet. (p) HANet. (q) SEIFNet. (r) Our LGCANet; black represents TN, white represents TP, red represents FN, and blue represents FP.

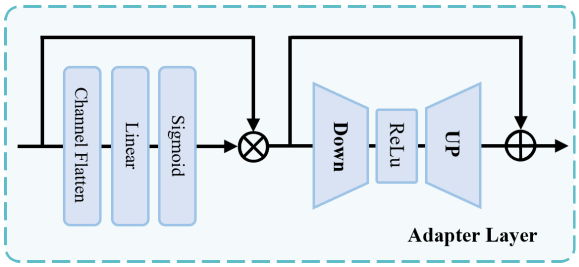


Fig. 10. Learnable adapter layer.

as illumination effects and seasonal changes. For irregular changed regions with different scales (Fig. 9, 4 and 5 rows), our network can identify the change boundaries accurately, which proves the effectiveness of designed EHM, which can enhance the model's edge awareness of changed regions.

D. Ablation Experiments and Analysis

1) Ablation Experiments: To prove the efficiency of the proposed components, we implemented ablation experiments on three CD tasks. Initially, we perform ablation experiments on the network framework and analyze the performance of each encoder utilization strategy. In addition, we introduced adapter versions for FastSAM for comparison. As shown in Fig. 10, we first use the channel dimension for compression, then use the sigmoid activation function to get the channel weights, which are multiplied with the input feature map, and the output is obtained after compression and recovery of the spatial dimension. As shown in Table IV, the model that uses ResNet encoder outperforms FastSAM on all datasets, which indicates that ResNet performs well in extracting local features and details. FastSAM performs well in global contextual information extraction, but lacks in detail capture of changed regions and has redundant features, thus affecting detection accuracy. After introducing the adapter, the model's performance unexpectedly declined. We attribute this to the following reasons. First, the limited capacity of the adapter may be insufficient to effectively capture the complex feature relationships, which restricts certain aspects of feature representation and leads to degraded performance. Additionally, when the downstream task has limited training data, the backbone model—pretrained on large-scale datasets—already possesses strong generalization ability. In contrast, the adapter, trained from scratch, is more likely to introduce task-irrelevant noise, potentially diluting or even disrupting the generalization knowledge learned by the backbone. Simply summing ResNet and FastSAM features gives a limited performance boost, but there will be redundancy of feature information, resulting in the learning of ineffective fusion information. ReSAM attains the best performance on all datasets, with its IoU improving by 1.29%, 0.67%, and 1.73% on all three tasks, respectively. This indicates that ReSAM fully utilizes the advantages of both encoders, enabling the model to understand the scene information more comprehensively and improve detection accuracy and robustness.

We further performed ablation experiments on the proposed modules, including CAAM, FAM, and EHM. The experimental results are illustrated in Table V. The performance of the network gradually increases with the sequential increase of the three modules, which demonstrates the effectiveness of each module. CAAM achieves local global feature extraction by capturing objects of various scales and shapes. FAM significantly increases the model's capability to recognize changed regions through feature interaction between changed and unchanged regions. EHM extracts edge information through orientation perception and generates fine-grained detection maps through feature selection mechanism. "ReSAM + CAAM + FAM" achieved the second better result, and compared to ReSAM, its IoU is improved by 2.57%, 3.23%, and 4.10% on the three datasets, respectively. Finally, the LGCANet incorporating all the modules obtains the best results on all datasets with IoU improvements of 2.74%, 3.62%, and 4.81%, respectively.

To verify the impact of the two parameters: offsets and weight in CAAM, we also performed hyperparameter ablation experiments on CAAM, and from the experimental results. As shown in Table VI, we can see that without offsets and weight, the $F1$ scores and IoU scores are lower on all three datasets. Using offsets and weight alone both improve the performance. Using both offsets and weight, $F1$ scores and IoU scores achieve the best results on all three datasets.

2) Feature Visualization: In Fig. 11, we illustrate the feature visualization results for various network phases. As seen in the figure, the features extracted earlier by ResNet and FastSAM

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TABLE IV
ABLATION STUDY ON DIFFERENT ENCODER STRATEGIES

Encoder	LEVIR-CD		BCDD		SYSU-CD	
	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
ResNet-18(Res)	86.72	81.29	89.81	87.84	78.11	64.09
FastSAM	85.07	79.89	87.96	86.18	77.12	62.76
FastSAM(Adapter)	**88.70**	79.70	91.05	83.58	77.24	62.92
Res+FastSAM	86.98	81.67	90.13	88.13	78.56	64.69
$\alpha \cdot$ Res+ $\beta \cdot$ FastSAM (ReSAM)	87.75	**82.58**	**93.59**	**88.51**	**79.38**	**65.82**

TABLE V
ABLATION STUDY ON DIFFERENT MODULES

Model Name	Module			LEVIR-CD		BCCD		SYSU-CD	
	CAAM	FAM	EHM	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
ReSAM	×	×	×	87.75	82.58	93.59	88.51	79.38	65.82
ReSAM+CAAM	✓	×	×	88.94	83.92	94.86	90.60	80.76	67.74
ReSAM+FAM	×	✓	×	88.66	83.51	94.99	90.82	80.40	67.23
ReSAM+EHM	×	×	✓	88.49	83.35	94.67	90.27	81.10	68.22
ReSAM+CAAM+FAM	✓	✓	×	89.90	85.15	95.53	91.74	82.29	69.92
ReSAM+CAAM+EHM	✓	×	✓	89.87	85.08	95.42	91.55	81.26	68.45
ReSAM+FAM+EHM	×	✓	✓	89.69	84.93	95.29	91.33	81.87	69.31
ReSAM+CAAM+FAM+EHM	✓	✓	✓	**89.78**	**85.32**	**95.76**	**92.13**	**82.79**	**70.63**

TABLE VI
ABLATION STUDY ON CAAM MODULE

Model Name	CAAM		LEVIR-CD		BCCD		SYSU-CD	
	Offsets	Weight	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
ReSAM+CAAM(V1)	×	×	88.22	83.21	94.38	89.80	79.97	66.63
ReSAM+CAAM(V2)	✓	×	88.41	83.68	94.64	90.23	80.62	67.54
ReSAM+CAAM(V3)	×	✓	88.57	83.72	94.74	90.41	80.58	67.48
ReSAM+CAAM	✓	✓	**88.94**	**83.92**	**94.86**	**90.60**	**80.76**	**67.74**

contain a lot of noise and not effectively focus on changed regions. After fusing features from both encoders, the network can achieve preliminary determination of changed regions. Later, three different modules also achieve positive effects for target feature capture, changed region extraction and boundary prediction. Fig. 11 further illustrates the efficiency of our presented module.

3) Computational Efficiency: Table VII shows the computational efficiency comparison of various methodologies on SYSU-CD dataset. In recently compared approaches, SRCDNet achieves fewer network parameter (Param) and floating-point operation per second (FLOPs), but it has limited performance. Our proposed network occupies 31.02M parameters, which is smaller than DSAMNet, ChangeFormer, DDCNN, and DETNet networks. The computational efficiency (FLOPs) of LGCANet is much lower than ChangeFormer, DDCNN, DSAMNet, and DETNet. Our network balances the parameter numbers and computational efficiency while achieving the best performance.

4) Failure Case Analysis: Fig. 12 presents failure cases on the SYSU-CD dataset, and it can be observed that the detection failures are attributed to two main reasons. First, the annotation of change boundaries in the SYSU-CD dataset is relatively broad, making the distinction between changed and unchanged areas complex. This is particularly challenging in edge regions or semantically ambiguous scenarios (such as vegetation cover changes or partial building renovations), where the model may struggle to accurately differentiate subtle changes from background noise, leading to increased missed detections or false alarms. Second, the model is overly

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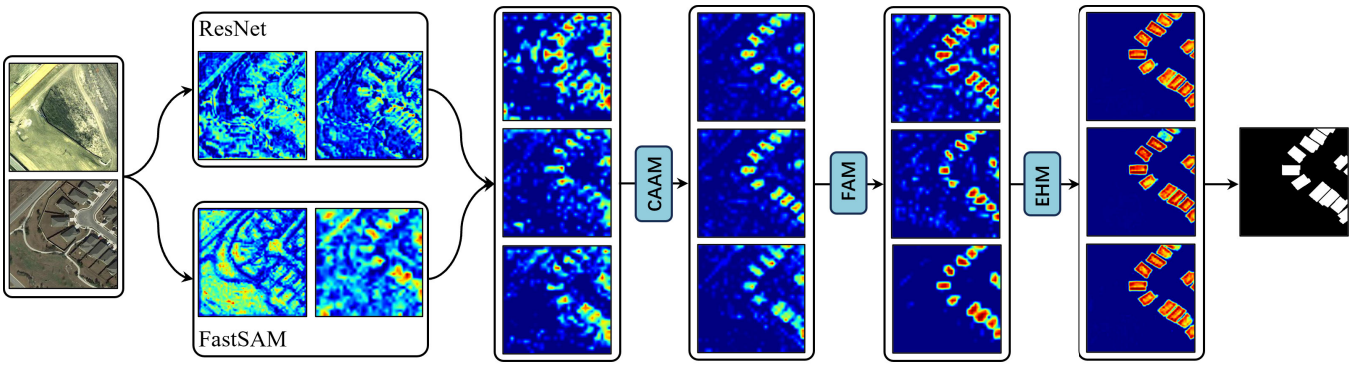


Fig. 11. Feature visualization of various network stages.

TABLE VII
SPATIAL COMPLEXITY AND PERFORMANCE ON SYSU-CD DATASET

Methods	Param.(M)	FLOPs.(G)	F1(%)	IoU(%)
FC-EF	1.34	3.56	75.24	60.30
FC-Diff	1.35	4.71	72.86	57.40
FC-Conc	1.54	5.32	76.43	61.92
HANet	2.61	17.62	73.98	58.71
BITNet	3.49	21.26	77.98	63.91
SNUNet	12.03	54.82	77.69	63.52
STANet	16.92	18.07	72.35	56.70
DSAMNet	16.95	75.38	75.03	60.04
SRCDNet	24.37	12.03	70.29	54.20
SEIFNet	27.90	8.36	81.75	69.13
UGSNet	32.09	26.67	81.31	68.50
ChangeFormer	41.01	202.62	76.27	64.60
DDCNN	46.67	135.78	77.06	62.74
DETNet	58.62	60.46	80.12	66.83
Ours	31.02	56.58	82.79	70.63



Fig. 12. Failure results on SYSU-CD datasets. (a) T1 moment images. (b) T2 moment images. (c) Label. (d) Ours; black represents TN, white represents TP, red represents FN, and blue represents FP.

sensitive to uninteresting changes, such as cars or containers being mistakenly detected as new buildings. This could be due to insufficient feature extraction by the model, which, in the absence of semantic information, incorrectly identifies these objects as building changes. To address these issues, we are considering incorporating object-level information (such as building and road categories) for change filtering to reduce false alarms on uninteresting changes. Additionally, we plan to introduce more samples of nontarget changes during the training process to enhance the model's ability to distinguish these changes.

V. CONCLUSION

Inefficient feature capture and effective discrimination of changed regions are the main problems of current deep learning-based CD algorithms. In this article, we present LGCANet via fast SAM, which models changed regions and semantic latent information for analyzing spatial-temporal differences. It utilizes ResNet and FastSAM for feature encoding and uses semantic information to constrain the learning of changed regions. Meanwhile, this article introduces CAAM to extract local–global complex semantic features, which are used to capture multiscale, various-shaped objects. Then, FAM is applied to optimize the semantic dissimilarity between changed features and unchanged features, which enhances the network's recognition of changed regions in a complementary manner. Finally, we implement EHM to extract the edge semantic features of changed regions, the interactions of different classes of features are enhanced by channel shuffle, and feature selection is performed to minimize the interference of nonchanged regions to further reinforce the robustness of detection results. Experiments prove that our proposed LGCANet superior to existing state-of-the-art approaches, which proves its outstanding performance in CD tasks.

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REFERENCES

[1] A. Singh, "Review article digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, 1989.
[2] S. Jin, L. Yang, P. Danielson, C. Homer, J. Fry, and G. Xian, "A comprehensive change detection method for updating the national land cover database to circa 2011," *Remote Sens. Environ.*, vol. 132, pp. 159–175, May 2013.
[3] D. Brunner, G. Lemoine, and L. Bruzzone, "Earthquake damage assessment of buildings using VHR optical and SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 5, pp. 2403–2420, May 2010.

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[4] C. N. Mundia and M. Aniya, "Analysis of land use/cover changes and urban expansion of Nairobi city using remote sensing and GIS," *Int. J. Remote Sens.*, vol. 26, no. 13, pp. 2831–2849, Jul. 2005.
[5] M. Hussain, D. Chen, A. Cheng, H. Wei, and D. Stanley, "Change detection from remotely sensed images: From pixel-based to object-based approaches," *ISPRS J. Photogramm. Remote Sens.*, vol. 80, pp. 91–106, Jun. 2013.
[6] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.
[7] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.
[8] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[9] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.
[10] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, no. 8, pp. 1–5, Aug. 2021.
[11] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.
[12] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Sep. 2020.
[13] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.
[14] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.
[15] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.
[16] X. Zhao et al., "Fast segment anything," 2023, *arXiv:2306.12156*.
[17] T. Wu, Y. Hu, J. Qin, X. Lin, and Y. Wan, "TPP-SAM: A trajectory points prompting segment anything model for zero-shot road extraction from high-resolution remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, no. 6, pp. 1–20, Jan. 2025.
[18] N. Liu, X. Xu, Y. Su, H. Zhang, and H.-C. Li, "PointSAM: Pointly-supervised segment anything model for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–15, 2025, 10.1109/TGRS.2025.3529031.
[19] L. Ding, K. Zhu, D. Peng, H. Tang, K. Yang, and L. Bruzzone, "Adapting segment anything model for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–11, 2024, doi: 10.1109/TGRS.2024.3368168.
[20] T. Zhou, Y. Zhao, and J. Wu, "ResNeXt and Res2Net structures for speaker verification," in *Proc. IEEE Spoken Lang. Technol. Workshop (SLT)*, Jan. 2021, pp. 301–307.
[21] B. Koonce, "EfficientNet," in *Convolutional Neural Networks With Swift for Tensorflow: Image Recognition and Dataset Categorization*. Springer, 2021, pp. 109–123.
[22] L. Yuan et al., "Tokens-to-token ViT: Training vision transformers from scratch on ImageNet," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2021, pp. 558–567.
[23] X. Zhai, A. Kolesnikov, N. Houlsby, and L. Beyer, "Scaling vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12104–12113.
[24] Y. Li, H. Fan, R. Hu, C. Feichtenhofer, and K. He, "Scaling language-image pre-training via masking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 23390–23400.
[25] X. Li et al., "OMG-seg: Is one model good enough for all segmentation?" in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27948–27959.
[26] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[27] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, pp. 2848–2864, 2020.
[28] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-resolution-based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2021.
[29] Z. Li, C. Yan, Y. Sun, and Q. Xin, "A densely attentive refinement network for change detection based on very-high-resolution bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022, doi: 10.1109/TGRS.2022.3159544.
[30] L. Wang, Z.-H. You, W. Lu, S.-B. Chen, J. Tang, and B. Luo, "Attention-aware Sobel graph convolutional network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, 2024, doi: 10.1109/TGRS.2024.3419790.
[31] Y.-C. Li, S. Lei, N. Liu, H.-C. Li, and Q. Du, "IDA-SiamNet: Interactive- and dynamic-aware Siamese network for building change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5628213.
[32] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," 2024, *arXiv:2404.03425*.
[33] Z. Wang, Y. Zhang, L. Luo, and N. Wang, "TransCD: Scene change detection via transformer-based architecture," *Opt. Exp.*, vol. 29, no. 25, pp. 41409–41427, 2021.
[34] C.-P. Chen, J.-W. Hsieh, P.-Y. Chen, Y.-K. Hsieh, and B.-S. Wang, "SARAS-net: Scale and relation aware Siamese network for change detection," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2023, vol. 37, no. 12, pp. 14187–14195.
[35] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2019, pp. 8024–8035.
[36] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[37] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5604816.
[38] D. He, Q. Shi, X. Liu, Y. Zhong, G. Xia, and L. Zhang, "Generating annual high resolution land cover products for 28 metropolises in China based on a deep super-resolution mapping network using Landsat imagery," *GISci. Remote Sens.*, vol. 59, no. 1, pp. 2036–2067, Dec. 2022.
[39] Q. Shi, M. Liu, A. Marinoni, and X. Liu, "UGS-1m: Fine-grained urban green space mapping of 34 major cities in China based on the deep learning framework," *Earth Syst. Sci. Data Discuss.*, pp. 1–23, Jun. 2022.
[40] C. Han, C. Wu, H. Guo, M. Hu, and H. Chen, "HANet: A hierarchical attention network for change detection with bitemporal very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3867–3878, 2023.
[41] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024, doi: 10.1109/TGRS.2024.3360516.



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